

PREPROCESSING AND DATA PREPARATION REPORT

Project Title: Used Car Price Prediction Engine

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1. PROBLEM STATEMENT

The goal of this project is to develop a machine learning regression model that predicts the market value of used cars based on their physical and operational characteristics (such as mileage, fuel type, transmission, and age). Providing an automated, objective valuation mechanism helps eliminate human appraisal errors and assists buyers and sellers in estimating fair market rates.

2. DATASET OVERVIEW

The project utilizes the CarDekho Used Car Dataset sourced from the Kaggle Machine Learning repository. The dataset contains 301 rows and 9 columns, with the target variable being `Selling_Price` (a continuous numerical variable representing price in lakhs).

This dataset was selected because it features a practical mix of continuous, discrete, and high-cardinality nominal variables. This layout requires a structured pipeline involving data cleaning, encoding, and feature scaling, making it highly suitable for demonstrating standard preprocessing steps.

3. PREPROCESSING STEPS IMPLEMENTED

The preprocessing pipeline was built sequentially to transform the raw data into a clean, machine-learning-ready format:

- **Deduplication:** Identified and permanently removed two duplicate rows to prevent artificial variance during model training.
- **Feature Engineering:** Calculated the age of each vehicle using the formula `Vehicle_Age = 2026 - Year` to turn the raw calendar year into an active numerical feature. Dropped the original `Year` column, along with `Car_Name` due to its high cardinality.
- **Outlier Mitigation:** Calculated the Interquartile Range (IQR) for continuous fields (`Kms_Driven` and `Present_Price`). Outliers lying beyond $1.5 \times \text{IQR}$ were capped to their upper and lower bounds to keep the dataset size intact while reducing data skew.
- **Categorical Encoding:** Applied One-Hot Encoding to nominal variables (`Fuel_Type`, `Seller_Type`, and `Transmission`). The first category of each variable was dropped to avoid multicollinearity (the dummy variable trap).
- **Data Splitting:** Partitioned the dataset into an 80% training set and a 20% testing set. This split was performed before any scaling or imputation to ensure no information from the test set could influence the preprocessing parameters (preventing data leakage).
- **Missing Value Imputation:** Set up a robust `SimpleImputer` using the median strategy. The imputer was fit on the training data and then used to transform both the train and test splits.

- **Feature Scaling:** Applied `StandardScaler` to all numerical features. This normalizes the data to have a mean of 0 and a standard deviation of 1, ensuring distance-sensitive algorithms do not heavily favor features with larger raw magnitudes.

4. CLASS IMBALANCE ANALYSIS

Because this is a regression task (predicting a continuous price) rather than a classification task, class balancing techniques such as SMOTE-Tomek are mathematically inapplicable. Class balancing was therefore bypassed with explicit technical justification.

5. TOOLS AND ENVIRONMENT

The environment and dependencies used for implementation include Python 3.10+, Pandas (for data manipulation), NumPy (for numerical operations), Scikit-learn (for splitting, imputation, and scaling), Google Colab, and Git/GitHub for version control and hosting.

6. TECHNICAL CHALLENGES AND RESOLUTIONS

The primary challenge was ensuring the pipeline remains resilient to future missing values without causing data leakage. This was resolved by using Scikit-learn's API to strictly run `.fit_transform()` on the training partition and only `.transform()` on the test partition. This guarantees the test set remains entirely unseen while maintaining a uniform pipeline structure.